

ECON 282E — Homework 1, Part B

From a Question to a Quantitative Answer: Method, Answer, Presentation

Foundations of Macroeconomics
University of California, Riverside

Out: Thursday, October 15, 2026 • Due (with Part A): Thursday, October 29, 2026

Where we are. In Part A you scoped the project, wrote the model down, validated a solver where the answer is known, and built your own data targets and calibration. Part B uses *your* calibration from A3(b) throughout. If you are not confident in it, you may use $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.025$, $\rho = 0.95$, $\sigma = 0.007$ instead — say so, and lose the A3(b) points only once.

Timing. B1 uses Session 4 (October 15). **B2(b)–(c) use Session 5 (October 22)**; everything else can be done as soon as this sheet is out.

What to hand in, together with Part A. The mini-paper (PDF), the five slides (PDF), the repository with `run_all.py`, the final charter, and `AI_LOG.md`. Question B2(d) feeds the anonymous class accuracy leaderboard.

B1. Choose the algorithm (15 points)

- (a) **(9 pts) What each tool buys, measured.** On the $\delta = 1$ model of A2 ($N = 200$), add to your grid VFI: Howard’s policy improvement; monotonicity of g to restrict the search; continuous choice by golden section (or Brent) on an interpolated value function; and *either* the endogenous grid method *or* time iteration on the Euler equation. Make one table: policy updates, wall time, $\max |g/g^* - 1|$, and the maximum unit-free Euler residual at 500 capital levels that are *not* grid points. Which of these changes the *answer*, and which only the *route* to it? Do your wall-clock times rank the way your operation counts do?
- (b) **(6 pts) Discretize the shock you estimated.** Implement Tauchen (1986) and Rouwenhorst (1995). At **your** ρ and σ from A3(b) and $n \in \{5, 7, 9, 15\}$, report each chain’s implied persistence and unconditional standard deviation against the targets. Which will you use, and why does your estimated ρ make the choice matter?

B2. Solve the calibrated model three ways (25 points)

Use the stochastic model of A1 at your calibration, a 7-state chain from B1(b), and capital on $[0.5k^*, 1.5k^*]$.

- (a) **(7 pts) A global solution.** Solve with your best solver from B1(a). Report the grid, tolerance, iterations and wall time, and check that the policy crosses the 45-degree line near the analytical k^* in the middle productivity state.
- (b) **(6 pts) First-order perturbation.** (*Session 5, Block A.*) Log-linearize, solve by QZ (or any method from 5.A4), and state the Blanchard–Kahn count you verified.

- (c) **(7 pts) Chebyshev collocation.** (*Session 5, Block B.*) Approximate consumption with a degree-7 Chebyshev polynomial in k for each productivity state, collocate the Euler residual at the Chebyshev nodes, and warm-start from (b).
- (d) **(5 pts) Accuracy against cost, and a decision.** One table for the three methods: numbers stored, solve time, and $\log_{10}$ of the maximum and mean unit-free Euler residual at 1,000 random off-grid points over the whole grid, and over $[0.95k^*, 1.05k^*]$. Submit your best maximum $\log_{10}$ residual, with its grid, tolerance and wall time, to the leaderboard. Then decide: **which method is fit for the positive question, and which for the normative one?** Be precise about the normative question: what does a first-order solution give if you also approximate *welfare* to first order (certainty equivalence, Session 5), and what if instead you evaluate exact utility along the first-order *policy*? Which is the right way to use it, and why?

B3. The answers (20 points)

- (a) **(10 pts) The positive answer.** Simulate the model, HP-filter the logs of Y , C and I , and put the six moments next to **your** data table from A3(a). Report them twice: from one long simulation, and as the mean and standard deviation across 200 simulated samples of the *same length as your data*. Which comparison is the right one, and why? Then answer the question: what fraction of the U.S. standard deviation of output (and of its variance) do TFP shocks account for in this model? Name the moment the model gets most wrong and the model feature most likely responsible. How much does switching to the Tauchen chain move your answer?
- (b) **(10 pts) The normative answer.** Define the welfare cost of fluctuations as the percentage λ by which consumption in every date and state of the stochastic economy must be raised to make the household as well off as in a deterministic economy where z is fixed at $\bar{z}$, both starting at the deterministic steady state $\bar{k}^*$:

$$\mathbb{E} \left[\sum_t \beta^t \ln((1 + \lambda)c_t) \right] = \frac{\ln \bar{c}^*}{1 - \beta}.$$

Compute λ from a global solution, averaging over the ergodic distribution of z , for two comparison economies: $\bar{z} = 1$, which preserves the mean of $\ln z$, and $\bar{z} = \mathbb{E}[z]$, which preserves the mean of z . Which comparison is fair, and why do the two differ? Compare your answer with Lucas's benchmark $\frac{1}{2}\text{var}(\ln c)$. Interpret the *size*: is it large or small, and what does that imply for stabilization policy in this model? Interpret the *sign*: could it be negative here, and what property of the technology would make it so? Finally, using your answer to A1(d), explain in one paragraph why an economy with uninsurable idiosyncratic risk can give a much larger number — the question Homework 2 takes up.

B4. Put it together (10 points)

- (a) **A mini-paper of at most six pages** (`templates/paper.tex`): question, model and equilibrium, calibration with its provenance, method and its accuracy, the two answers, and what the model cannot say. Every number in it must appear in the output of `run_all.py`.
- (b) **Five slides** (`templates/slides.tex`): the question, the model, the evidence that the solution is right, the answer, the caveat. Write them as if you had eight minutes.

- (c) **A replication package.** `python run_all.py` downloads or reads the data, solves, simulates and writes every table and figure. Pin your package versions.
- (d) **AI_LOG.md with a validation ledger:** for each claim in the paper, the check that supports it (a closed form, a refinement, an Euler residual, a second method, a verified source). Add a half-page retrospective: what broke, what you would do differently, and which charter risk actually happened.

Graded on correctness, clarity and reproducibility — not on length.

B5. Your own question (10 points)

On one page, pitch a question *you* would like to answer, as a project manager would. State the question and whether it is positive or normative; the model you would build on and why a simpler one will not do; the equilibrium concept; the solution method you would choose from Sessions 3–5 and the accuracy check you would run; and the data (structured or not) that would discipline it. It may become your final project (teams are due November 5). Name the component of *this* homework — your calibration and data pipeline, a solver, your route — that the project would reuse, and what it would still need from Homework 2. Post the pitch to the *Pitches and teams* category of the course forum (under your name or anonymously) so that teams can form around it.

B6. Choose your route (20 points)

Research projects differ in where their risk lies. Pick **one** route and go deeper; the route you choose is a component your final project can reuse. Say in your charter which route you chose and why.

R1. The numerical frontier (Sessions 4–5).

Solve the calibrated model by **second-order perturbation** and recompute the welfare cost of B3(b) with it. How close is it to the welfare cost from the first-order and the global *policies*, each evaluated by exact utility? Explain the size of the differences you find (think envelope theorem). Simulate the second-order solution with and without **pruning** and report whether it matters at your calibration. Then either raise risk (triple σ) until perturbation visibly fails, or solve the endogenous-labour model of the bonus on a two-dimensional **Smolyak** grid or with finite elements.

R2. Data and estimation (Sessions 3–4).

Replace the calibration of ρ and σ by **simulated method of moments**: choose at least three moments from your A3(a) table, a weighting matrix, and a simulation size, and *first* show that your estimator recovers known parameters from data simulated by the model. Then estimate on the U.S. moments, report standard errors, and test the overidentifying restriction. Does the model fit what you asked it to fit?

R3. Policy (Session 2).

Add a proportional tax τ_k on capital income, rebated lump sum. Derive the steady-state effect of τ_k on k^* and c^* , and state the Ramsey steady-state tax in this model (Chamley–Judd). Compute the welfare effect, *including the transition*, of a permanent surprise move from $\tau_k = 0$ to $\tau_k = 0.25$ starting at the old steady state. Explain why the steady-state comparison and the transition-inclusive answer differ, and what a government without commitment (Session 2, deck 2.B3) would be tempted to do.

Bonus (up to 10 points). Endogenous labour

Add a labour choice with $u(c, n) = \ln c - \psi \frac{n^{1+1/\nu}}{1+1/\nu}$ and calibrate ψ so that average hours are $\bar{n} = 0.3$. How do $\sigma(Y)$, $\sigma(I)/\sigma(Y)$ and $\sigma(H)/\sigma(Y)$ change, and does your answer to B3(a) go up or down?

Grading at a glance. Part A 100 points, Part B 100 points (B1 15, B2 25, B3 20, B4 10, B5 10, B6 20), bonus 10. What earns the points is a decision you can defend and a number you can reproduce and have checked. A correct number with no evidence that it is correct earns little; a well-diagnosed failure earns a lot.